

Operations on sets

Definition: subset

Let A and B be sets. A is a subset of B if every element of A are also elements of B ($A \subset B$).

e. g. $A = \{1, 2, 3\}$, $B = \{1, 2, 3, 4\}$

$$\Rightarrow A \subset B$$

$\subset$ means subset and $\subseteq$ means subset and/or equal

2^n is the number of subsets in a finite set with cardinality n .

Definition : Equality of sets

Let A and B be sets. $A = B$ if $A \subseteq B$ and $B \subseteq A$.

Operations on sets :

$A \cap B$ means the intersection of both sets A and B , $\{x : x \in A \wedge x \in B\}$

$A \cup B$ means the union of A and B $\{x : x \in A \vee x \in B\}$

$A \setminus B$ means the difference between A and B , $\{x : x \in A \wedge x \notin B\}$

Definition : Complement of a set

Given all of the sets being considered are all subsets of a fixed set S , we call S the **universal set** and define the complement of set A as:

$$A^c = A' = \{x \in S : x \notin A\} = S \setminus A$$

Definition: Powerset

A set of all of the subsets of a given set S is called the **powerset**, denoted using $P(S)$.

$$|S| = n \Rightarrow P(S) = 2^n$$

Definition: Cartesian Product

For sets A and B , their cartesian product $(A \times B) = \{(a, b) : a \in A \wedge b \in B\}$

It consists of all the ordered pairs (a, b) with $a \in A$ and $b \in B$

Consider $A \times B = \mathbb{R}$

$$\begin{aligned}\mathbb{R} \times \mathbb{R} &\stackrel{?}{=} \mathbb{R} \\ &= \{(x, y) : \\ &\quad x \in \mathbb{R} \wedge y \in \mathbb{R}\}\end{aligned}$$

= set of cartesian
pairs in 2-dimensional
space

Proving set operation equalities

1) Recall definitions

2)

3) Relate properties of set A to set B